



ICT Department Website Redesign: Smart & Interactive Web Application

Project Documentation

Project Title: Smart and Interactive ICT Department Website

Subject: Internet Programming 2 (ITPR300/ITNP300)

Assessment: Assessment 2

Submission Date: 13 May 2026

Institution: Department of Information and Communication Technology (ICT)

Qualification: Diploma: Information Technology (IT)

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1. Introduction

1.1 Project Overview

This project involves the redesign and development of the current ICT Department website into a smart, interactive, and data-driven web application. The goal is to move beyond a static informational site and create a dynamic platform that serves students, staff, and prospective stakeholders.

1.2 Objectives

- Full-stack web application using HTML, CSS, JavaScript, Node.js, Express, and database
- Apply UXD principles
- Interactive frontend features
- RESTful backend
- Database integration (MySQL)
- Middleware implementation
- AI-driven feature (chatbot)
- Responsive design

2. Website Analysis & Problem Identification

Key Issues:

- Complex navigation (High)
- Poor accessibility (High)
- Cluttered layout (Medium)
- Static content (High)
- Not responsive (Critical)
- No interactivity (Critical)
- No database (Critical)

Proposed Improvements:

- Navigation redesign
- Responsive layout
- Accessibility improvements
- Interactivity features
- Database integration
- AI chatbot

3. UXD & Frontend

Principles:

- User-centered design
- Consistency
- Accessibility
- Readability
- Responsiveness
- Feedback

Pages:

- Home, Programs, Events, Bursaries, Timetables, Login

4. Frontend Interactivity

Features:

- Mobile menu
- Form validation
- Filtering
- Modals
- Feedback messages
- Search
- Loading spinners

5. Backend (Node.js & Express)

- Express server
- API routes (GET, POST, PUT, DELETE)
- Example: enquiry submission

6. Database (MySQL)

Tables:

- Users
- Events
- Announcements
- Enquiries
- Staff
- FAQs

CRUD operations implemented.

7. Middleware

Types:

- Logging
- Authentication
- Validation
- Error handling

8. AI Feature

Chatbot:

- Answers FAQs
- Uses keyword matching
- Helps users instantly

9. Integration & Testing

- Frontend + Backend + Database connected
- All test cases passed

10. GitHub & Code Quality

- Organized structure
- README included
- Clean coding practices

11. Presentation Plan

- 8-minute structured presentation
- Made a video through teams App

12. Team Contributions

Each member contributed to:

- Frontend
- Backend
- Database
- Testing
- Documentation

13. Appendices

Includes:

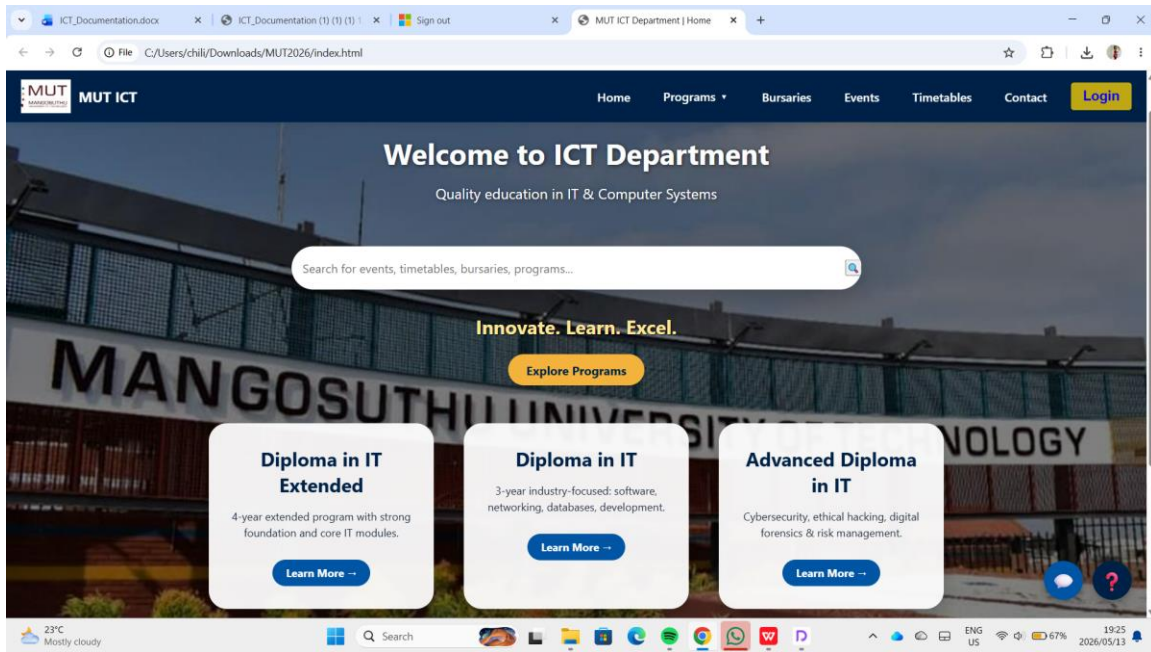
- Setup instructions
- Code snippets
- Resources

Conclusion

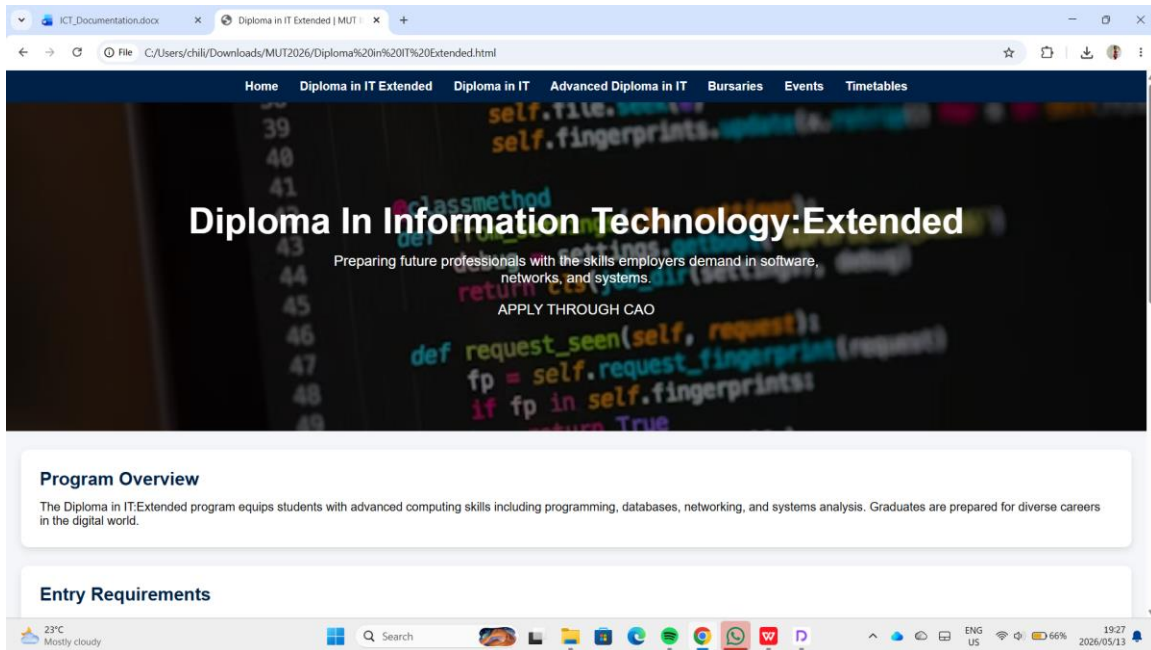
The project successfully transforms the ICT website into a modern, interactive, and intelligent platform meeting all requirements.

Pages

1. Home page



2. Programs Page



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Diploma in IT Extended | MUT

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Entry Requirements

- National Senior Certificate or equivalent
- English First Additional Language(3)OR
- English Home Language(3)AND
- Mathematics(2)OR
- Mathematics Literacy(3)
- Atleast 50% in three fundamental subjects including English
- Atleast 60% in three compulsory vocational subjects

Duration Of Study

- 4 years full-time

Career Paths

- Software Developer
- Systems Analyst
- Network Engineer
- IT Consultant
- Database Administrator

Key Skills You Will Gain

Programming

Master languages like Java, Python, and C# to build real-world software.

Networking

Learn how to design, configure, and secure modern networks.

Database Management

Understand database systems, SQL, and data analysis techniques.

23°C Mostly cloudy

Search

ENG US

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Student Spotlight

"The Diploma In IT:Extended program gave me the confidence and skills to start my career in software development. The labs and projects were amazing!"

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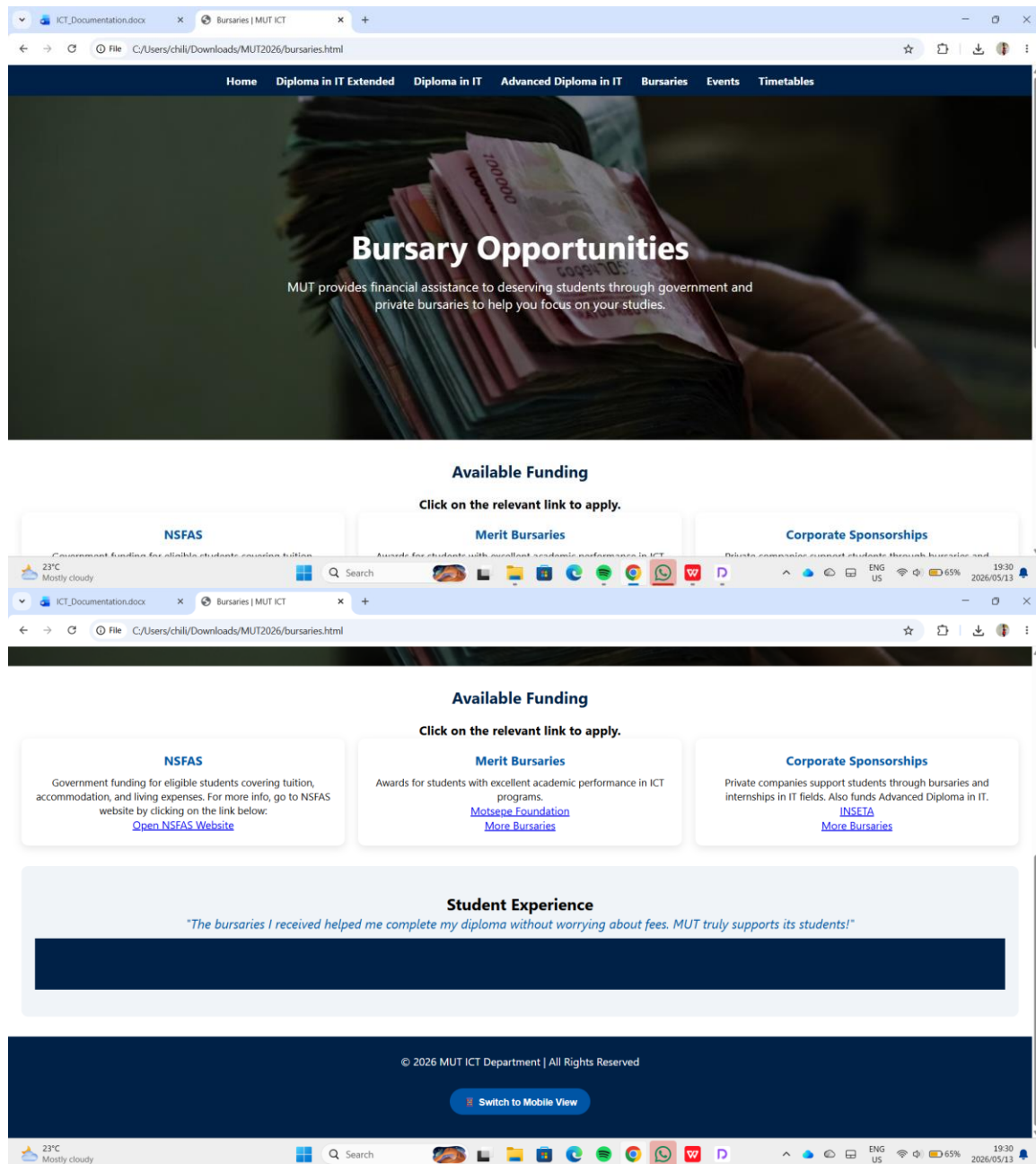
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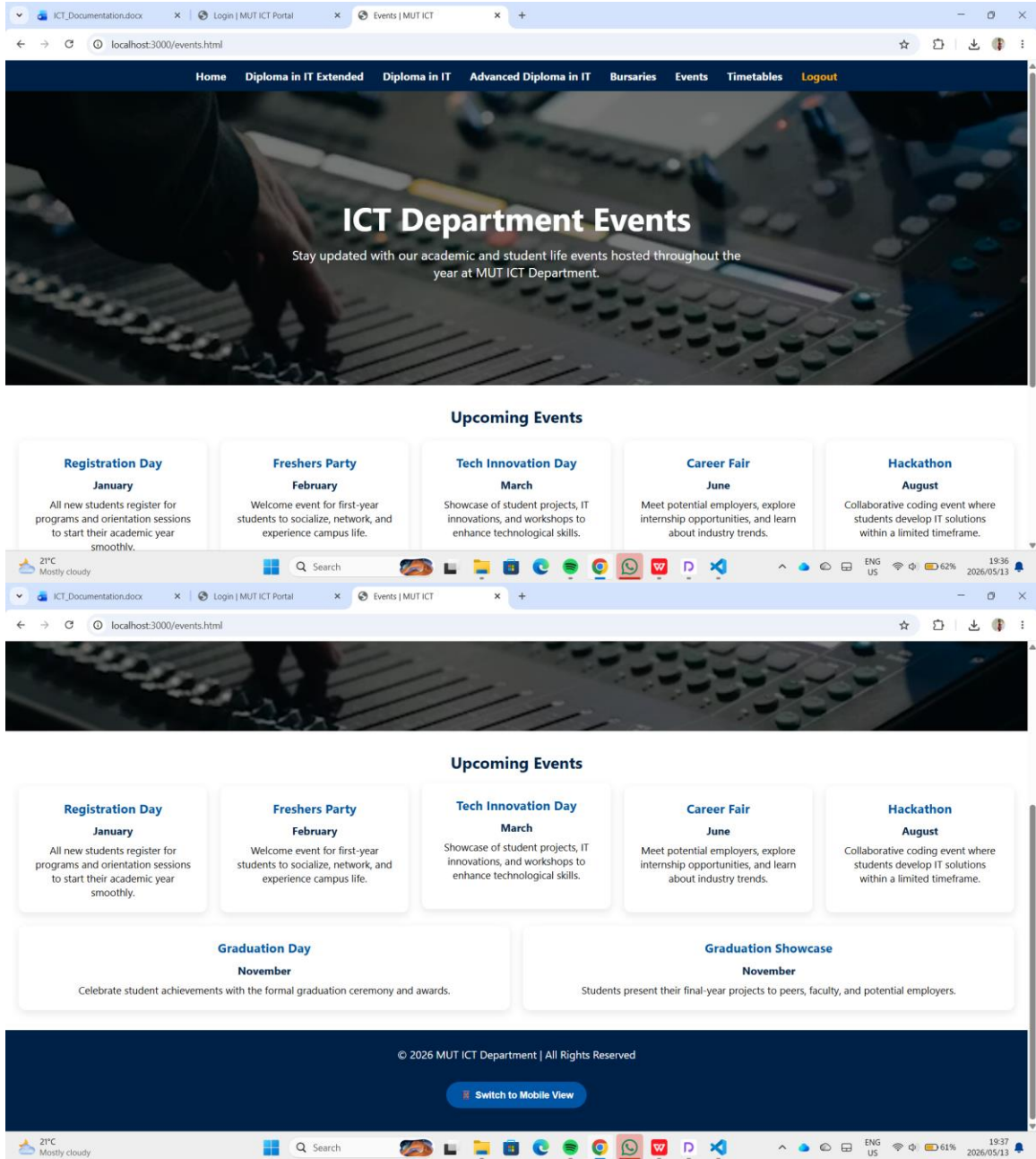
ENG US

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3. Bursaries Page



4. Events Page



The screenshot shows a web browser displaying the MUT ICT Timetables page. The page has a dark blue header with the MUT ICT logo and navigation links: Home, Programs, Bursaries, Events, Timetables, Contact, and Logout. The main content area is white with a blue section header for 'First Year – IT (Extended & Core)'. Below this, it says '2026 Academic Year | All Groups'. A table titled 'IT First Year 2024' displays the academic schedule for various days and time slots.

IT First Year 2024	07:45 to 09:00	09:15 to 10:40	11:00 to 12:25	13:05 to 14:25	14:30
MONDAY	DS1 – NS1	TP1 – NS1	SS1 – NS1		
TUESDAY	IS1 – LH14				
WEDNESDAY		DS1 – NS1	TP1 – NS1	SS1 – LH11	IS1 – D9
THURSDAY		IS1 – L3	DS1 – NS1	SS1 – NS1	TP1 – NS1
FRIDAY					

Effective: 2 February 2026 | Version 1.0



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YEAR2: CN TIMETABLE

Group Time Table, Version 1.0

Effective: 2 Feb 2026

CN2	1	2	3	4		5	6		7	8	9	10
	07:45 08:25	08:30 09:10	09:15 09:55	10:00 10:40		11:00 11:40	11:45 12:25		13:05 13:45	13:50 14:30	14:35 15:10	15:15 16:00
MON			ITSKILLS NWS	CNET2 NWS	B R E A K	TUT	TUT	B R E A K	DIS2 L22	ITEL2 L22		
TUES			ITEL2 R14	ITEL2 R14		CNET2 R14	CNET2 R14		ITSKILLS	ITSKILLS	DIS2 R14	DIS2 R14
WED			ITEL2 L7	CNET2 L7		TUT	TUT		DIS2 L4	TUT	TUT	IT SKILL L13
THURS			ITEL2 R14	ITEL2 R14		DIS2 R14	DIS2 R14		CNET2 R14	CNET2 R14	OLT	ITSKILLS OLT
FRI				Testing		Testing	TUT					

Department of Information and Communication Technology

SD8	1	2	3	4	5	6	7	8	9	10
	07:45	08:30	09:15	10:00	11:00	11:45	13:05	13:50	14:35	15:15
	08:25	09:10	09:55	10:40	11:40	12:25	13:45	14:30	15:10	16:00

6. Login Page

